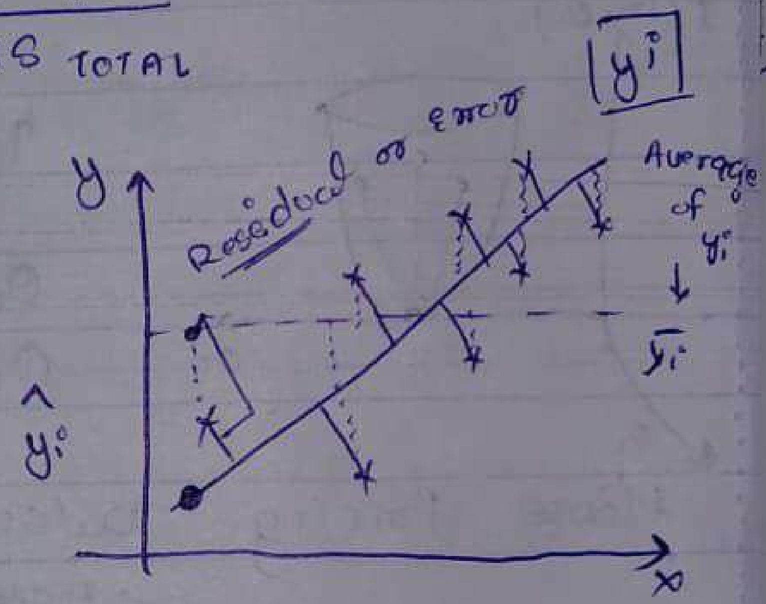
∴ SS → sum of square

Performance Metrics Used In linear Regression

1) R squared 2) Adjusted R squared

$$R \text{ squared} = 1 - \frac{SS_{Res}}{SS_{TOTAL}}$$

$$= 1 - \frac{\Sigma(y_i - \hat{y}_i)^2}{\Sigma(y_i - \bar{y}_i)^2}$$



$$= 1 - \frac{\text{small number}}{\text{Bigger number}}$$

$= 1 - $ small number

≈ 1 → $\boxed{1}$

↓ Accurate my model is?

0.70 ⇒ 70 %

0.85 ⇒ 85 %

0.90 ⇒ 90 %

{Overfitting, Underfitting}

R squared ↑↑↑

2) Adjusted R squared

Size of the house ↑ Price ↑

$\boxed{\text{+ve correlation}}$

Dataset → (Price)

~~Size of the house~~ ~~Price~~

Gender

Size of the house	No. of bedrooms	Location

↔ Price

No. of bedroom ↑ Price ↑

$\boxed{\text{+ve correlation}}$

R squared => 75% => 0.75

R squared => 80% => 0.80

R squared => 85% 0.85

R squared => 87% 0.87

Note: This is the problem of R square even though you don't have a input feature with is directly corelated with output feature then also the value will increases.

for resolve this we use Adjusted R squared

$$\text{Adjusted R squared} = 1 - \frac{(1-R^2)(N-1)}{N-p-1}$$

N = No. of data points

p = No. of Independent feature

$$\begin{cases} p = 2 & R^2 = 90\% & R^2\text{adjusted} = 86\% \\ p = 3 & R^2 = 92\% & R^2\text{adjusted} = 82\% \end{cases}$$

Note: if the additional independent feature is directly corelated with output feature the Adjusted R squared value will increas with previous ones, if not then decrease with previous ones.

(In the case of addition independent feature is not directly corelated with output feature)